

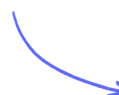
Calculator Questions

Introduction to Algebra

Algebraic Notation / Algebraic Vocabulary / Substitution / Collecting Like Terms

Easy (4 questions)	/7
Medium (8 questions)	/18
Hard (5 questions)	/13
Total Marks	/38

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Easy Questions

- 1 Find the value of $7x + 3y$ when $x = 12$ and $y = -6$.

(2 marks)

2 $f = 3g + 7h$

Work out the value of f when $g = -5$ and $h = 2$

(2 marks)

- 3 Sofia is trying to simplify $\frac{6c + 10}{2}$

Her method is
divide $6c$ by 2
then
add 10

Evaluate her method.

(1 mark)

- 4 Use the formula $s = ut + \frac{1}{2}at^2$.

Calculate s when $u = 5$, $t = 10$ and $a = 3$.

$s = \dots\dots\dots$

(2 marks)

Medium Questions

1 $s = ut + \frac{1}{2}at^2$

Find the value of s when $u = 5.2$, $t = 7$ and $a = 1.6$.

$s =$

(2 marks)

2 $y = mx + c$.

Find the value of y when $m = -2$, $x = -7$ and $c = -3$.

$y =$

(2 marks)

- 3 (a) The body mass index, B , for a person of mass m kg and height h metres is given by the formula

$$B = \frac{m}{h^2}$$

Usman has a mass of 50 kg.

He has a height of 1.57 m.

Work out Usman's body mass index.

Give your answer correct to one decimal place.

(2 marks)

- (b) Tom's height is 1.80m. He wants his body mass index to be 21

Work out the mass that will give Tom a body mass index of 21.

(2 marks)

$$A = 4bc$$

4

$$A = 100$$

$$b = 2$$

Work out the value of c .

(2 marks)

5 $x = 0.7$

Work out the value of $\frac{(x+1)^2}{2x}$

Write down all the figures on your calculator display.

(2 marks)

6 $w = 5y^2 - y^3$

Work out the value of w when $y = -2$

$w =$

(2 marks)

7 $G = c^2 - 4c$

Find the value of G when $c = -5$

$G =$

(2 marks)

8 Find the value of s when $u = 12$, $a = 10$ and $t = 4$.

$$s = ut + \frac{1}{2}at^2$$

(2 marks)

Hard Questions

1 $y = \sqrt{u^2 x}$

Find the value of y when $u = 7$ and $x = 25$.

$y = \dots\dots\dots$

(2 marks)

2 $s = ut + \frac{1}{2}at^2$

Find s when $t = 26.5$, $u = 104.3$, $a = -2.2$.

Give your answer in standard form, correct to 4 significant figures.

$s = \dots\dots\dots$

(4 marks)

3 $\frac{a}{b} = 3c$

$$\frac{b}{c} = 2$$

Work out the value of a when $c = 8$

(3 marks)

4 a and b are positive values.

Show that $\frac{7a+2b-3a}{8a+6b+2a-b}$ always simplifies to the same value.

(3 marks)

5 Amy has x beads.

Billy has three more beads than Amy.

Carly has four times as many beads as Billy.

Circle the expression for the number of beads that Carly has.

$4x + 3$	$3x + 4$	$4(x + 3)$	$x + 12$
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(1 mark)